



NATIONAL OCCUPATIONAL STANDARD FOR LIQUEFIED PETROLEUM GAS INSTALLER

APPROVING AUTHORITY

This National Occupational Standard has been prepared and published under the authority of the Zambia Qualifications Authority Board on 21st September 2012.

ZAMBIA QUALIFICATIONS AUTHORITY

The Zambia Qualifications Authority Act No. 13 of 2011 was enacted by the Government of the Republic of Zambia to ***“provide for the development and implementation of a national qualifications framework; establish the Zambia Qualifications Authority; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing”***. Among other functions, ZAQA is responsible for ***determining national standards for any occupation***, through various sector specific National Occupational Standards Development Teams (NOSDTs).

REVISION OF NATIONAL OCCUPATIONAL STANDARDS

National Occupational Standards shall be revised every after **5 years**, or whenever necessary, by the issue of either amendments or of revised editions. It is important that users of National Occupational Standards (NOS) should ascertain that they are in possession of the latest amendments or editions.

NOS DEVELOPMENT TEAM RESPONSIBLE

This National Occupational Standard was prepared by the Energy National Occupational Standards Development Team, upon which the following organisations were represented:

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2. Department of Energy - Ministry of Energy;
3. Energy Regulation Board;
4. Kafue Gorge Regional Training Centre;
5. Mines Safety Department - Ministry of Mines and Minerals Development;
6. Ministry of Labour and Social Security Services;
7. Muhanya Solar;
8. Northern Technical College;
9. Rural Electrification Authority;
10. SNV Zambia;
11. University of Zambia;
12. Zambia Qualifications Authority.

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ACRONYMS AND ABBREVIATIONS

CS	Core Skill
LPG	Liquefied Petroleum Gas
LPGI	Liquefied Petroleum Gas Installer
NOS	National Occupational Standard
NOSDT	National Occupational Standards Development Team
OK	Organisational Knowledge
PC	Performance Criteria
PS	Professional Skill
RK	Regulatory Knowledge
RPL	Recognition of Prior Learning
TK	Technical Knowledge
ZAQA	Zambia Qualifications Authority
ZQF	Zambia Qualifications Framework

GLOSSARY OF TERMS

For the purposes of this NOS, the following terms and definitions shall apply:

Core Skills/Generic Skills: are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.

Function: is an activity necessary for achieving the key purpose of the sector, occupation, or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of NOS.

Job Title: defines a unique set of functions that together form a unique employment opportunity in an organisation.

Knowledge and Understanding: are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

National Occupational Standards (NOS): are statements of the standards of performance individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding. They are precise descriptions of what an individual is expected to be able to do in his/her work role.

National Occupational Standards (NOS) Code: is a unique reference code that identifies a NOS.

National Occupational Standards Development Team (NOSDT): means an established group of national stakeholders/experts responsible for the development of National Occupational Standards within a specific economic sector or occupation.

Occupation: is a set of job roles, which perform similar/related set of functions in an industry.

Organisational Context: includes the way the organisation is structured and how it operates, including the extent of operative knowledge that managers have in their relevant areas of responsibility.

Performance Criteria: are statements that together specify the standard of performance required when carrying out a task.

Scope: is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.

Sector: is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.

Sub Sector: is derived from a further breakdown based on the characteristics and interests of its components.

Technical Knowledge: is the specific knowledge needed to accomplish specific designated responsibilities.

Unit Title: gives a clear overall statement about what the incumbent should be able to do.

1. OVERVIEW

This is an introductory section providing a brief summary and specific information or commentary about the content of the NOS and the targeted sector and occupation to help the user judge whether it is relevant to them.

NOS Code	NOS.LPGI.01
Occupation	System Design, Installation, Operation and Maintenance, Testing and Commissioning
Job Title	LPG System Installer
Job Description	LPG System Installers are responsible for system designing, installing, operating, maintaining and testing and commissioning of LPG systems, and ensure that they meet the performance and reliability needs of customers by complying with all applicable codes, standards, procedures, specifications and safety requirements.
Job Purpose	The Individual on the job selects, sizes, installs, inspects, tests, operates and maintains, tests and commissions gas lines and gas equipment such as pressure vessels, meters and regulators in residential, commercial and industrial establishments.
ZQF Level	Level 6
Sector	Energy
Sub sector	All subsectors of the Energy sector
Other Economic Sector(s) in which the Occupation is Practiced	Construction, Transport, Hospitality, Academia, Research and Development, Waste Management, Regulatory and Standards Agencies
Other Similar Jobs that can be performed by the LPG System Installer	• Quality assurance • Energy audit • Compliance
Minimum Educational Job Entry Qualification(s)	Level 6
Practicing License Requirements (if any)	Engineering Registration Board Practicing Licence
Training/RPL	• Internship in LPG sub sector • Training in instrumentation and process control
Minimum Job Entry Age	21 years
Prior Experience	At least 1 year relevant experience in the energy sector, inclusive of relevant internship
Performance Criteria	As described in the Units under Section 4

2. SCOPE

This National Occupational Standard specifies the fundamental knowledge and understanding, skills and competences that an LPG System Installer must possess to be successful in his/her job role.

3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)

This job requires an individual to possess:

- Reading Skills
- Writing Skills
- Oral Communication Skills
- Creativity
- Problem Solving Skills
- Analytical Skills
- Mathematical Skills
- Integrity
- Interpersonal Skills
- Commercial Awareness
- Attention to Detail

4. UNITS AND ELEMENTS

This National Occupational Standard is divided into five (05) Units representing the tasks that a job holder should undertake in his/her day to day work. Each unit is further broken down into elements depicting the number of activities to be carried out for the successful execution of a particular task.

UNIT 1 [This unit is about health, safety and environmental protection requirements for an LPG System Installer to be proficient in designing, installing, operating, maintain, testing and commissioning of LPG systems safely].

Unit No.	01
Unit Title	Safety, health and environmental management
Description	This unit is about maintaining safety, health and environmental protection for the individual(s) and the plant and equipment.
Scope	This unit covers the following: • Ensure all health and safety regulations are followed by the incumbent as well as other employees. • Ensure the operations of the establishment conform to the requirements for environmental protection. • Ensure the safety of personnel, plant and equipment.
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Ensure all Health and safety regulations are followed by the incumbent as well as other employees	To be competent, the individual must be able to: PC1. Read, interpret and implement the safety and health policies of the organisation. PC2. Read, interpret and implement the safety and health regulations for the organisation. PC3. Identify hazards and assess the risks of all aspects of operations for plant and equipment.
Ensure the operations of the establishment conform to the requirements for environmental protection.	To be competent, the individual must be able to: PC4. Read, interpret and implement the environmental policies of the organisation. PC5. Read, interpret and implement the environmental regulations of the organisation. PC6. Identify environmental hazards and assess risks of all aspects of operations for plant and equipment.
Ensure the safety of plant and equipment	To be competent, the individual must be able to: PC7. Read, interpret and implement the standard operating procedures for plant and equipment of the organisation. PC8. Identify hazards and assess risks of all aspects of operations for plant and equipment.
Knowledge and Understanding (K)	
A. Organisation-al Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company health and safety policies and regulations. OK2. Company environmental policies and regulations. OK3. Company standard operating procedures.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of:

	TK1. Health and Safety risk assessment. TK2. Environmental Risk assessment. TK3. Plant and equipment operational risk assessment.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Government regulatory agency requirements for health and safety. RK2. Government regulatory agency requirements for environmental protection.
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills
	The individual on the job must be able to: CS1. Read and interpret internal/external documents. CS2. Read and understand manuals, health and safety instructions, memos, other company documents. CS3. Ability to read from different sources- books, screens in machines and signage.
	Writing Skills
	The individual on the job must be able to: CS4. Fill up documentation applicable to one's role. CS5. Write minutes of meetings. CS6. Write technical reports, budgets, and other relevant documents.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS7. Express statements or information clearly so that others can hear and understand. CS8. Participate in and understand the main points of discussions. CS9. Respond appropriately to any queries.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision making process. PS2. Take decisions with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet set targets. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow code of conduct. PS6. Manage relationships with customers with intent on satisfying requirements for service delivery.
	Problem Solving
	The individual on the job must be able to: PS7. Recognize problems and search for solutions. PS8. Choose appropriate methods to complete assigned tasks. PS9. Approach relevant authority when required.
	Analytical Thinking

	The individual on the job must be able to: PS10. Apply risk based approach to analyse situations for action. PS11. Apply domain knowledge, observations and data to select appropriate course of action to perform tasks.
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from various sources to perform day to day activities. PS13. Ask questions for better understanding to evaluate the complexity of the task and seek assistance and support whenever required.

UNIT 2 [This unit is about system design requirements for an LPG System Installer to be proficient in designing of LPG systems safely].

Unit No.	02
Unit Title	Prepare system designs for LPG installation
Description	This unit is about sizing, specifications and preparation of drawings and cost estimates
Scope	This unit covers the following: • Sizing equipment, piping and regulation devices. • Specifying technical details and work methods for installing plant and equipment. • Preparation of Installation drawings for LPG systems. • Preparation of cost estimates based on prevailing market conditions.
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Sizing equipment, piping and regulation devices for LPG systems	To be competent, the individual must be able to: PC1. Size equipment based on applicable standards and regulatory requirements. PC2. Size piping based on applicable standards and regulatory requirements. PC3. Size regulation devices based on applicable standards and regulatory requirements.
Specifying technical details and work methods for installing plant and equipment for LPG systems	To be competent, the individual must be able to: PC4. Specify technical details for plant and equipment, piping and regulation devices. PC5. Specify work methods and tooling for installing plant and equipment, piping and regulation devices.
Preparation of Installation drawings for LPG systems	To be competent, the individual must be able to: PC6. Prepare schematic diagrams and process flow diagrams. PC7. Prepare detailed materials and equipment data sheets.
Preparation of cost estimates based on prevailing market conditions for LPG systems	To be competent, the individual must be able to: PC8. Prepare detailed bills of materials and quantities. PC9. Source the market for up to date cost estimates for the materials, works and services. PC10. Prepare a comprehensive priced bills of materials and quantities.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company standard operating procedures. OK2. Company rules, culture and practices. OK3. Company vision, mission and values. OK4. Company strategic direction and objectives/targets.

B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Plant and equipment operational risk assessment. TK2. Existing standards and guidelines for specifying equipment, piping and regulation devices. TK3. Industry practices, work methods and specific tooling. TK4. Preparation of schematic diagrams and process flow diagrams using relevant tools and software. TK5. Structure and detail of specification for bills of quantities.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Government regulatory agency requirements for health and safety. RK2. Government regulatory agency requirements for environmental protection. RK3. Government regulatory agency requirements for energy.
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills
	The individual on the job must be able to: CS1. Read and interpret internal/external documents. CS2. Read and interpret manuals, health and safety instructions, memos, other company documents. CS3. Read and interpret various safety and general signage, safety rules and tags etc. provided in the work place, including directions for exit routes during emergencies.
	Writing Skills
	The individual on the job must be able to: CS4. Fill up documentation applicable to one's role. CS5. Write minutes of meetings. CS6. Write technical reports, budgets, and other relevant documents
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS7. Express statements or information clearly so that others can hear and understand. CS8. Participate in and understand the main points of discussions. CS9. Respond appropriately to any queries
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision making process. PS2. Take decisions with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet set targets. PS4. Work constructively and collaboratively with others.

	Customer Centricity
	The individual on the job must be able to: PS5. Follow code of conduct. PS6. Manage relationships with customers with intent on satisfying requirements for service delivery.
	Problem Solving
	The individual on the job must be able to: PS7. Recognize problems and search for solutions. PS8. Choose appropriate methods to complete assigned tasks. PS9. Approach relevant authority when required.
	Analytical Thinking
	The individual on the job must be able to: PS10. Apply risk based approach to analyse situations for action PS11. Apply domain knowledge, observations and data to select course of action to perform tasks related to LPG installations.
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from various sources to perform day to day activities. PS13. Ask questions for better understanding to evaluate the complexity of the task and seek assistance and support whenever required.

UNIT 3 [This unit is about system installation requirements for an LPG System Installer to be proficient in installing of LPG systems safely].

Unit No.	03
Unit Title	Install LPG plant and equipment
Description	This unit is about reading and interpreting installation drawings and relevant specifications, applying correct methods to install plant and equipment and performing quality checks on the installed plant and equipment.
Scope	This unit covers the following: • Reading and interpreting installation drawings and relevant specifications. • applying correct methods to install plant and equipment. • performing quality checks on the installed plant and equipment.
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Reading and interpreting installation	To be competent, the individual must be able to: PC1. Read and interpret installation drawings specifications. PC2. Read and interpret specifications. PC3. Read and interpreting work methods.
Applying correct methods to install plant and equipment	To be competent, the individual must be able to: PC4. Mark and set out layout for plant installation. PC5. Check and inspect plant and equipment before installation. PC6. Install piping, plant and equipment and regulation devices. PC7. Install labels using appropriate colour codes and materials. PC8. Prepare As-built drawings.
Performing quality checks on the installed plant and equipment	To be competent, the individual must be able to: PC9. Carry out pressure testing on piping network. PC10. Ensure functional operation of all supporting works including electrical, civil and mechanical works. PC11. Ascertain correct operation of regulation devices. PC12. Ascertain correct labelling and use of correct colour coding.
Knowledge and Understanding (K)	
A. Organisation-al Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company standard operating procedures. OK2. Company rules, culture and practices. OK3. Company vision, mission and values. OK4. Company strategic direction and objectives/targets.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Plant and equipment operational risk assessment. TK2. Existing standards and guidelines for installing equipment, piping and regulation devices. TK3. Industry practices, work methods and specific tooling.

	TK4. Reading and interpreting of schematic diagrams and process flow diagrams. TK5. Use of relevant measuring instruments, tooling and test equipment. TK6. Labelling and colour coding for LPG installations.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Government regulatory agency requirements for health and safety. RK2. Government regulatory agency requirements for environmental protection. RK3. Government regulatory agency requirements for energy.
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills
	The individual on the job must be able to: CS1. Read and interpret internal/external documents. CS2. Read and understand manuals, health and safety instructions, memos, other company documents. CS3. Read and interpret various safety and general signage, safety rules and tags etc. provided in the work place, including directions for exit routes during emergencies.
	Writing Skills
	The individual on the job must be able to: CS4. Fill up documentation applicable to one's role. CS5. Write minutes of meetings. CS6. Write technical reports, budgets, and other relevant documents.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS7. Express statements or information clearly so that others can hear and understand. CS8. Participate in and understand the main points of discussions. CS9. Respond appropriately to any queries.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision making process. PS2. Take decisions with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet set targets. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow code of conduct. PS6. Manage relationships with customers with intent on satisfying requirements for service delivery.

	Problem Solving
	The individual on the job must be able to: PS7. Recognize problems and search for solutions. PS8. Choose appropriate methods to complete assigned tasks. PS9. Approach relevant authority when required.
	Analytical Thinking
	The individual on the job must be able to: PS10. Apply risk based approach to analyse situations for action. PS11. Apply domain knowledge, observations and data to select course of action to perform tasks related to LPG installation.
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from various sources to perform day to day activities. PS13. Ask questions for better understanding to evaluate the complexity of the task and seek assistance and support whenever required.

UNIT 4 [This unit is about operation and maintenance requirements for an LPG System Installer to be proficient in operating and maintaining LPG systems safely].

Unit No.	04
Unit Title	Operate and maintain LPG systems
Description	This unit is about applying standard operating procedures, readings instruments and gauges, estimating consumption of LPG, and conducting routine checks, preventive and corrective maintenance.
Scope	This unit covers the following: • Application of standard operating procedures. • Reading instruments and gauges, observing and reporting conditions. • Estimating consumption of LPG. • Conducting routine checks, and preventive and corrective maintenance.
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Application of standard operating procedures	To be competent, the individual must be able to: PC1. Read, interpret and implement standard operating procedures. PC2. Identify hazards and assess the risks of all aspects of operations and maintenance for plant and equipment.
Reading instruments and gauges, observing and reporting conditions	To be competent, the individual must be able to: PC3. Read and record measurements from instruments and gauges. PC4. Observe and interpret the operating conditions of the plant and equipment. PC5. Report operational status of plant and equipment as necessary.
Estimating consumption of LPG	To be competent, the individual must be able to: PC6. Calculate consumption rates for LPG. PC7. Make comparison on LPG consumption rates to industry standards.
Conducting routine checks, and preventive and corrective maintenance	To be competent, the individual must be able to: PC8. Carry out routine checks on plant and equipment. PC9. Trend recorded measurements and conditions to determine abnormal operating conditions for action. PC10. Carryout troubleshooting and diagnostics on plant and equipment. PC11. Plan maintenance schedules based on equipment supplier recommendations as well as industry practices. PC12. Carry out maintenance through service and repair of plant and equipment as necessary.

Knowledge and Understanding (K)	
A. Organisation-al Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company standard operating procedures. OK2. Company rules, culture and practices. OK3. Company vision, mission and values. OK4. Company strategic direction and objectives/targets.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Industry standards and practices. TK2. Instrumentation for LPG process flow control. TK3. Maintenance, philosophy, strategies and programs.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Government regulatory agency requirements for health and safety. RK2. Government regulatory agency requirements for environmental protection. RK3. Government regulatory agency requirements for energy.
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills
	The individual on the job must be able to: CS1. Read and interpret internal/external documents. CS2. Read and understand manuals, health and safety instructions, memos, other company documents. CS3. Read and interpret various safety and general signage, safety rules and tags etc. provided in the work place, including directions for exit routes during emergencies.
	Writing Skills
	The individual on the job must be able to: CS4. Fill up documentation applicable to one's role. CS5. Write minutes of meetings. CS6. Write technical reports, budgets, and other relevant documents.
B. Professional Skills	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS7. Express statements or information clearly so that others can hear and understand. CS8. Participate in and understand the main points of discussions. CS9. Respond appropriately to any queries.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision making process. PS2. Take decisions with systematic course of actions and/or response.

	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet set targets. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow code of conduct. PS6. Manage relationships with customers with intent on satisfying requirements for service delivery.
	Problem Solving
	The individual on the job must be able to: PS7. Recognize problems and search for solutions. PS8. Choose appropriate methods to complete assigned tasks. PS9. Approach relevant authority when required.
	Analytical Thinking
	The individual on the job must be able to: PS10. Apply risk based approach to analyse situations for action PS11. Apply domain knowledge, observations and data to select course of action to perform tasks related to LPG installation.
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from various sources to perform day to day activities. PS13. Ask questions for better understanding to evaluate the complexity of the task and seek assistance and support whenever required.

UNIT 5 [This unit is about testing and commissioning requirements for an LPG System Installer to be proficient in testing and commissioning of LPG systems safely].

Unit No.	05
Unit Title	Test and Commission LPG systems
Description	This unit is about testing and commissioning the plant and equipment
Scope	This unit covers the following: • Verifying proper functioning of the equipment/system after installation; • Verifying that the performance of the installed equipment/systems meets the specified design intent through a series of tests and adjustments; and • Capturing and recording performance data of the whole installation as the baseline for future operation and maintenance.
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Verifying proper functioning of the equipment/ system after installation	To be competent, the individual must be able to: PC1. Prepare the test and commissioning procedure. PC2. Confirm compliance to statutory requirements. PC3. Inspect and confirm the as-built LPG system to the design layout. PC4. Test the process flow parameters e.g. pressure, flow rate and temperature. PC5. Carry out leak detection checks on the LPG system.
Verifying that the performance of the installed equipment/ systems meets the specified design intent through a series of tests and adjustments	To be competent, the individual must be able to: PC6. Test and compare process flow parameters for the as-built LPG system to design parameters and industry standards. PC7. Adjust plant, equipment and regulation devices to meet performance levels.
Capturing and recording performance data of the whole installation as the baseline for future operation and maintenance	To be competent, the individual must be able to: PC8. Prepare a testing and commissioning report for LPG installations. PC9. Issue testing and commissioning certificate for LPG installations.

Knowledge and Understanding (K)	
A. Organisation-al Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company standard operating procedures. OK2. Company rules, culture and practices. OK3. Company vision, mission and values. OK4. Company strategic direction and objectives/targets.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Testing and commissioning check lists for LPG installations. TK2. LPG plant, equipment and regulation devices. TK3. Industry practices and performance benchmarks for LPG systems.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Government regulatory agency requirements for health and safety. RK2. Government regulatory agency requirements for environmental protection. RK3. Government regulatory agency requirements for energy.
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills
	The individual on the job must be able to: CS1. Read and interpret internal/external documents. CS2. Read and understand manuals, health and safety instructions, memos, other company documents. CS3. Read and interpret various safety and general signage, safety rules and tags etc. provided in the work place, including directions for exit routes during emergencies.
	Writing Skills
	The individual on the job must be able to CS4. Fill up documentation applicable to one's role. CS5. Write minutes of meetings. CS6. Write technical reports, budgets, and other relevant documents.
B. Professional Skills	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS7. Express statements or information clearly so that others can hear and understand. CS8. Participate in and understand the main points of discussions. CS9. Respond appropriately to any queries.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision making process. PS2. Take decisions with systematic course of actions and/or response.

	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet set targets. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow code of conduct. PS6. Manage relationships with customers with intent on satisfying requirements for service delivery.
	Problem Solving
	The individual on the job must be able to: PS7. Recognize problems and search for solutions. PS8. Choose appropriate methods to complete assigned tasks. PS9. Approach relevant authority when required.
	Analytical Thinking
	The individual on the job must be able to: PS10. Apply risk based approach to analyse situations for action. PS11. Apply domain knowledge, observations and data to select course of action to perform tasks related to LPG installations.
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from various sources to perform day to day activities. PS13. Ask questions for better understanding to evaluate the complexity of the task and seek assistance and support whenever required.

5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS

These include, but not limited to;

1. Computer with application software appropriate for LPG systems.
2. Personal Protective Equipment (PPE) appropriate for LPG environments.
3. Measuring equipment and tools appropriate for installation, operation and maintenance, and testing and commissioning of LPG plant and equipment.

6. DILEMMAS/CHALLENGES AND COMPLEXITIES FOR A JOB HOLDER

Dilemmas associated with the job of a chemical engineer include working in dangerous areas and hazardous machinery/equipment, working in confined areas, long working hours, exposure to chemical, physical and biological hazards, pressure from supervisors and colleagues, time pressure to complete tasks, pressure from government regulators, working in extreme weather such as hot and cold conditions, working in noisy, wet and dusty environments, etc.

6.1 Alternative Choices (Solutions) to Dilemmas and Complexities

Solutions to dilemmas include wearing protective clothing (PPE) and ensuring their availability and use by other employees, exercising regularly to maintain physical fitness, participating in workplace safety sensitisation and awareness meetings/training sessions, planning your daily tasks, reading and researching new ways to handle work stress, adhering to company's safety and standard operating procedures at all times, consulting extensively within and outside one's department/team on safety and other issues.

7. WORKING CONDITIONS/ENVIRONMENT

Working conditions include working in cold, hot and wet conditions, working at heights, stand/walk for long hours, working in laboratory environment, working in shifts, areas that are noisy and dusty, areas with limited lighting and ventilation. Working in confined spaces, working in explosives environment.

8. PARTIES INVOLVED/INTERACTING WITH THE JOB HOLDER OR TRAINEE

8.1 Internal/Within the Organization

Parties involved/interacting with the job holder who are internal to the organization include supervisors, subordinates, and other employees (e.g. architects, surveyors, engineers, etc).

8.2 External/Outside the Organisation

Parties involved/interacting with the job holder who are external include customers/clients, trainers, government regulators, suppliers of equipment/tools/consumables, fellow LPG System Installers from other companies, labour unions/occupational health and safety associations, etc.

9. PHYSICAL DEMANDS ON THE BODY

- Physique to sustain strenuous conditions;
- Walk and stand for long periods of time;
- Bend, stretch, twist, or reach out;
- Lift relatively heavy materials, tools and equipment;
- Use fingers, hands and feet with ease to complete the assigned task (dexterity).

ANNEX A

Criteria for Assessments based on this NOS

A.1 Guidelines for Assessment

- A.1.1** Criteria for assessment for curricula and learning programmes based on this NOS will be created by curricula and programmes developers. Each Performance Criteria (PC) will be assigned marks proportional to its importance in the NOS. Curricula and programmes developers will also lay down proportion of marks for theory and practical skills for each performance criteria, giving more weight to practical skills.

There shall be allocated the 'Total Mark', which will be the sum of all marks in each Unit, distributed across the number of PCs in that particular Unit. The 'out of' mark will be the mark allocated to each PC, which will be shared between theory and skills practical assessments.

- A.1.2** Awarding/assessment bodies or institutions and other users of the NOS will create unique question papers for the theory part and evaluations for skill practical part for their respective candidates.

ANNEX B NOS Version Control

This Annex gives details necessary for the tracking of the NOS versions based on the number of revisions.

NOS Code	NOS.LPGI.01		
ZQF Level	Level 6	Version Number	01
Sector	Energy	Date of Approval	September 2021
Sub Sector	All subsectors of the Energy sector	Date of Last Review	N/A
Occupation	LPG System Installer	Date of Next Review	September 2026

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